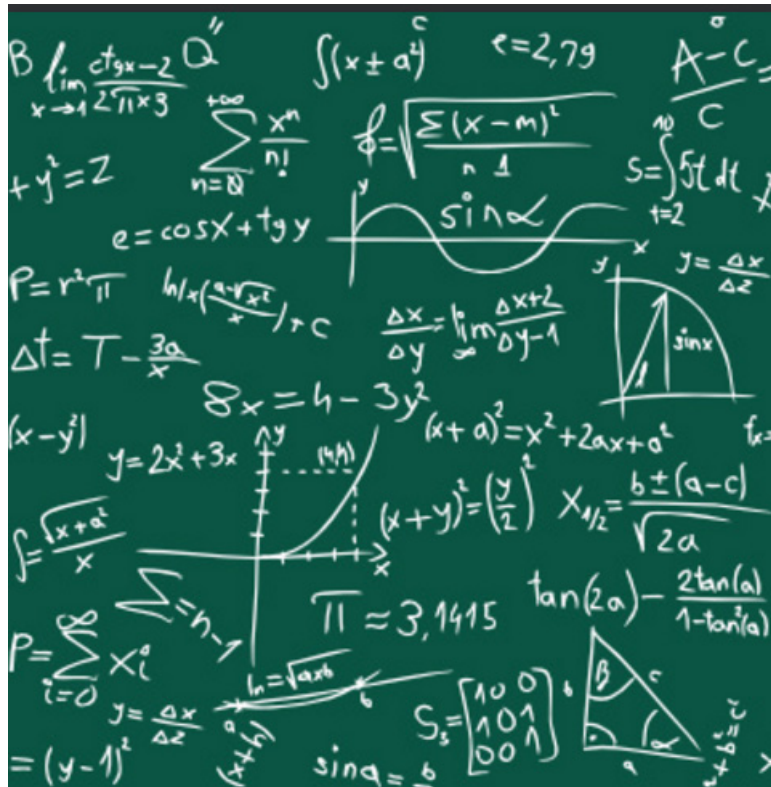


Exercises: Bisection method



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Using the bisection method, find the solutions of the equation

$$x^3 + 2x - 1 = 0$$

correct to two decimal places.

(First of all, one must determine how many solutions there are.)

Written in the form

$$x^3 = 1 - 2x,$$

the equation suggests first comparing the graphs of the functions $y = x^3$ and $y = 1 - 2x$, in order to determine beforehand the number of solutions of the equation. Comparing the two graphs, we deduce that there exists a unique solution $\alpha \in (0, 1)$, i.e., a unique zero of the function $f(x) = x^3 + 2x - 1$ in that interval. More formally:

$$\exists! \alpha \in (0, 1) : f(\alpha) = 0$$

We therefore proceed by applying the bisection method to the interval $[a, b] = [0, 1]$, with $f(0) = -1 < 0$ and $f(1) = 2 > 0$.

Iteration 1 ($n = 0$)

$$[a_0, b_0] = [0, 1]$$

$$c_0 = \frac{a_0 + b_0}{2} = \frac{1}{2}$$

with $f(c_0) = \frac{1}{8} > 0$. Then, in the next iteration, $b_1 = c_0$.

Iteration 2 ($n = 1$)

$$[a_1, b_1] = \left[0, \frac{1}{2}\right], f(0) < 0, f\left(\frac{1}{2}\right) > 0$$

$$c_1 = \frac{a_1 + b_1}{2} = \frac{1}{4}$$

with $f(c_1) < 0$. Then, in the next iteration, $a_2 = c_1$.

Iteration 3 ($n = 2$)

$$[a_2, b_2] = \left[\frac{1}{4}, \frac{1}{2}\right], f\left(\frac{1}{4}\right) < 0, f\left(\frac{1}{2}\right) > 0$$

$$c_2 = \frac{a_2 + b_2}{2} = \frac{3}{8}$$

with $f(c_2) < 0$. Then, in the next iteration, $a_3 = c_2$. To reach the desired level of approximation, we will have to wait until iteration 9 ($n = 8$), where we will obtain

$$a_8 = \frac{29}{64} \simeq 0,453 \quad b_8 = \frac{117}{256} \simeq 0,457$$

We can finally state that $\alpha \in \left(\frac{29}{64}, \frac{117}{256}\right)$ correct to two decimal places.

Exercise 2

Using the bisection method, find the solutions of the equation

$$x + \log x = 0$$

correct to two decimal places.

(First of all, one must determine how many solutions there are.)

Written in the form

$$\log x = -x,$$

the equation suggests first comparing the graphs of the functions $y = \log x$ and $y = -x$, in order to determine beforehand the number of solutions of the equation. Comparing the two graphs, we deduce that there exists a unique solution $\alpha \in (0, 1)$, i.e., a unique zero of the function $f(x) = x + \log x$ in that interval. More formally:

$$\exists! \alpha \in (0, 1) : f(\alpha) = 0$$

We therefore proceed by applying the bisection method to the interval $[a, b] = [0 + \varepsilon, 1]$, $\varepsilon > 0$, with $f(1) = 1 > 0$ and $f < 0$ in a right neighborhood of the point $x_0 = 0$ of arbitrarily small radius ε (note that $y = \log x$ is not defined at $x = 0$).

Iteration 1 ($n = 0$)

$$\begin{aligned} [a_0, b_0] &= [0 + \varepsilon, 1] \\ c_0 &= \frac{a_0 + b_0}{2} = \frac{1 + \varepsilon}{2} \simeq \frac{1}{2} \end{aligned}$$

with $f(c_0) = \frac{1}{2} - \log 2 < 0$. Then, in the next iteration, $a_1 = c_0$.

Iteration 2 ($n = 1$)

$$\begin{aligned} [a_1, b_1] &= \left[\frac{1}{2}, 1 \right], f\left(\frac{1}{2}\right) < 0, f(1) > 0 \\ c_1 &= \frac{a_1 + b_1}{2} = \frac{3}{4} \end{aligned}$$

with $f(c_1) > 0$. Then, in the next iteration, $b_2 = c_1$.

Iteration 3 ($n = 2$)

$$\begin{aligned} [a_2, b_2] &= \left[\frac{1}{2}, \frac{3}{4} \right], f\left(\frac{1}{2}\right) < 0, f\left(\frac{3}{4}\right) > 0 \\ c_2 &= \frac{a_2 + b_2}{2} = \frac{5}{8} \end{aligned}$$

with $f(c_2) > 0$. Then, in the next iteration, $b_3 = c_2$.

Iteration 4 ($n = 3$)

$$[a_3, b_3] = \left[\frac{1}{2}, \frac{5}{8} \right], f\left(\frac{1}{2}\right) < 0, f\left(\frac{5}{8}\right) > 0$$

$$c_3 = \frac{a_3 + b_3}{2} = \frac{9}{16}$$

with $f(c_3) < 0$. Then, in the next iteration, $a_4 = c_3$. Observe that

$$a_3 = 0,5, b_3 = 0,625$$

and we are therefore still far from having two correct decimal places. To reach the desired level of approximation, we will have to wait until iteration 10 ($n = 9$), where we will obtain

$$a_9 = \frac{145}{256} \simeq 0,566 \quad b_9 = \frac{291}{512} \simeq 0,568$$

We can finally state that $\alpha \in \left(\frac{145}{256}, \frac{291}{512}\right)$ correct to two decimal places.

Exercise 3

Using the bisection method, find the solutions of the equation

$$2 \sin x = x$$

correct to one decimal place.

(First of all, one must determine how many solutions there are.)

Exercise 4

Using the bisection method, find the solutions of the equation

$$x^2 - 2 - \log x = 0$$

correct to one decimal place.

(First of all, one must determine how many solutions there are.)